SZPIRO'S CONJECTURE AND THE ABC CONJECTURE

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This worksheet is a short version of [Sil09, §VIII.11]. The minimal discriminant Δ_E of an elliptic curve E over $\mathbb{Q}$ and its conductor N_E are divisible by the same primes, the primes of bad reduction. In N_E they appear to small powers, while in Δ_E a prime may appear to a large power. Szpiro's conjecture bounds $|\Delta_E|$ by a constant times $N_E^{6+\varepsilon}$. It would already imply Fermat's last theorem for all sufficiently large exponents [Sil09, Proposition VIII.11.2]. It is closely related to the ABC conjecture, which bounds the height of a rational point $[A : B : C]$ of the line $X+Y = Z$ by a power of the radical of ABC. Problem 1.5 computes, with the help of `Magma`, the minimal discriminant and the reduction types of the curve $y^2 = x(x + A)(x - B)$ attached to $A + B = C$. Problems 2.6 and 3.2 use it to show that each conjecture implies a form of the other. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$. A Weierstrass equation for E has invariants c_4, c_6 and discriminant Δ [Sil09, §III.1], which lie in $\mathbb{Z}$ when its coefficients do.
- A prime p, with valuation v_p on $\mathbb{Q}$, and the ring $\mathbb{Z}_{(p)}$ of rational numbers whose denominators are prime to p.

How to use this. Allow about 25 minutes for Problem 1.5, 15 for Problem 2.6 and 20 for Problem 3.2. `Magma` is needed only in Problem 1.5, and its blocks run in seconds in the free online calculator. Everything else is by hand.

1. The minimal discriminant of a Frey curve

Problem 1.5 proves [Sil09, Lemma VIII.11.3]. Silverman bounds the scaling factor u of a change of variables with two identities that write $288A^2$ and $288B^2$ in terms of c_4 and c_6. Here `Magma` finds identities in terms of c_4 and Δ instead, and they give both parts of the lemma at once.

Definition 1.1 (Minimal equations [Sil09, §§VII.1, VII.5])**.** A Weierstrass equation for E is minimal at p if its coefficients lie in $\mathbb{Z}_{(p)}$ and $v_p(\Delta)$ is as small as possible among such equations. It is global minimal if its coefficients lie in $\mathbb{Z}$ and it is minimal at every prime p. The curve E has good, multiplicative or additive reduction at p if the reduction modulo p of an equation minimal at p is nonsingular, has a node, or has a cusp.

Proposition 1.2 (Reduction types [Sil09, Remark VII.1.1 and Proposition VII.5.1])**.** *Consider a Weierstrass equation for E with coefficients in $\mathbb{Z}_{(p)}$.*

(1) If $v_p(\Delta) < 12$ or $v_p(c_4) < 4$, then the equation is minimal at p.
(2) Suppose that the equation is minimal at p. Then E has good reduction at p if $v_p(\Delta) = 0$, multiplicative reduction if $v_p(\Delta) > 0$ and $v_p(c_4) = 0$, and additive reduction if $v_p(\Delta) > 0$ and $v_p(c_4) > 0$.

Definition 1.3 (Minimal discriminant [Sil09, §VIII.11])**.** The curve E has a global minimal Weierstrass equation [Sil09, Corollary VIII.8.3]. Its discriminant is the minimal discriminant Δ_E of E. It

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does not depend on the equation, since two global minimal equations are related by a change of variables with $u = \pm 1$ [Sil09, Proposition VII.1.3(b)].

Proposition 1.4 (Changes of variables [Sil09, Proposition III.3.1(b) and Table 3.1])**.** *Any two Weierstrass equations for an elliptic curve over $\mathbb{Q}$ are related by a change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ with $u \in \mathbb{Q}^\times$ and $r, s, t \in \mathbb{Q}$. If the first has discriminant Δ and invariant c_4, then the second has discriminant $u^{-12}\Delta$ and invariant $u^{-4}c_4$.*

In this section, A, B, C are nonzero integers with $A + B = C$ and $\gcd(A, B, C) = 1$, and E is the elliptic curve

$$E\colon y^2 = x(x + A)(x - B). \tag{1.1}$$

The invariant c_4 and the discriminant Δ are those of this equation.

Problem 1.5 (The minimal discriminant of a Frey curve)**.**
The following code works in the polynomial ring $\mathbb{Z}[A, B]$. It checks that $c_4 = 16F$ and $\Delta = 16G^2$ for the polynomials F and G below, and writes A^3 and B^3 as combinations of F and G with coefficients in $\mathbb{Z}[A, B]$.

```
P<A, B> := PolynomialRing(Integers(), 2);
// y^2 = x(x + A)(x - B), over the field of fractions of P
E := EllipticCurve([FieldOfFractions(P) | 0, A - B, 0, -A*B, 0]);
c4 := P!cInvariants(E)[1];
Delta := P!Discriminant(E);
F := A^2 + A*B + B^2;
G := A*B*(A + B);
c4 eq 16*F, Delta eq 16*G^2;      // true true
I := IdealWithFixedBasis([F, G]);
Coordinates(I, A^3);              // [ A, -1 ]
Coordinates(I, B^3);              // [ B, -1 ]
```

(1) Show that A, B, C are pairwise coprime, and that $\gcd(c_4, \Delta) = 16$.[a]
(2) Show that the minimal discriminant of E satisfies $|\Delta_E| = 2^4(ABC)^2$ or $|\Delta_E| = 2^{-8}(ABC)^2$. In particular, $|\Delta_E| \geqslant 2^{-8}(ABC)^2$.[b]
(3) Show that E has multiplicative reduction at every odd prime p dividing ABC.

[a] *Hint:* a prime dividing $F(A, B)$ and $G(A, B)$ divides A^3 and B^3.
[b] *Hint:* let a global minimal equation come from (1.1) by a change of variables with scaling factor u, as in Proposition 1.4. Show that $u \in \mathbb{Z}$ and that u^4 divides $\gcd(c_4, \Delta)$.

2. From Szpiro to ABC

Problem 2.6 is the proof of [Sil09, Proposition VIII.11.5(a)]. It applies Conjecture 2.4 to the curve (1.1) of Problem 1.5, whose minimal discriminant is large and whose conductor is small. Szpiro made his conjecture in 1983, and Masser and Oesterlé proposed the ABC conjecture in 1985. Silverman's displayed definition of the conductor takes $f_p(E) = 2$ at every prime of additive reduction, which he notes is not quite correct at 2 and 3. At these two primes the correct exponent has a technical definition, sketched in Remark 2.2. The two conductors differ by a factor dividing $2^6 \cdot 3^3$, which the constants in the conjectures absorb.

Definition 2.1 (Conductor [Sil09, §VIII.11])**.** The conductor of E is

$$N_E \coloneqq \prod_{\substack{p \\ 2}} p^{f_p(E)},$$

where the exponent $f_p(E)$ is 0 if E has good reduction at p, 1 if it has multiplicative reduction at p, and 2 if it has additive reduction at p and $p \geqslant 5$. If E has additive reduction at $p = 2$ or $p = 3$, then $f_p(E)$ is defined in [Sil94, §IV.10], and $2 \leqslant f_2(E) \leqslant 8$ and $2 \leqslant f_3(E) \leqslant 5$.

Remark 2.2 (The exponents f_2 and f_3 [Sil09, §C.16])**.** Let p be a prime at which E has additive reduction. The inertia group at p acts on the Tate modules of E, and an integer $\delta_p(E) \geqslant 0$ measures the wild ramification in this action. It vanishes for $p \geqslant 5$, and $f_p(E) = 2 + \delta_p(E)$.

The exponent can also be computed directly. For any prime p, let m_p be the number of irreducible components, ignoring multiplicities, of the special fiber of the minimal complete Néron model of E at p. Then the Ogg–Saito formula [Sil09, Proposition C.16.2] gives

$$f_p(E) = v_p(\Delta_E) + 1 - m_p.$$

Definition 2.3 (Height and radical)**.** Every rational point P of the line $X + Y = Z$ in $\mathbb{P}^2$ has coordinates $P = [A : B : C]$ with $A, B, C \in \mathbb{Z}$ and $\gcd(A, B, C) = 1$, unique up to sign. Then $A + B = C$. The height of P is

$$H(P) \coloneqq \max\bigl(|A|, |B|, |C|\bigr).$$

The radical of a nonzero integer n is

$$R(n) \coloneqq \prod_{p \mid n} p.$$

Conjecture 2.4 (Szpiro [Sil09, Conjecture VIII.11.1])**.** *For every $\varepsilon > 0$ there is a constant κ_ε such that every elliptic curve E over $\mathbb{Q}$ satisfies*

$$|\Delta_E| \leqslant \kappa_\varepsilon N_E^{6+\varepsilon}.$$

Conjecture 2.5 (ABC [Sil09, Conjecture VIII.11.4])**.** *For every $\varepsilon > 0$ there is a constant λ_ε with the following property. Let $P = [A : B : C]$ be a rational point of the line $X + Y = Z$ in $\mathbb{P}^2$, with $A, B, C \in \mathbb{Z}$ coprime and $ABC \neq 0$. Then*

$$H(P) \leqslant \lambda_\varepsilon R(ABC)^{1+\varepsilon}.$$

Problem 2.6 (Szpiro implies ABC with exponent 3/2)**.**
Assume Conjecture 2.4. Let $P = [A : B : C]$ be a rational point of the line $X + Y = Z$ in $\mathbb{P}^2$, with $A, B, C \in \mathbb{Z}$ coprime and $ABC \neq 0$. You will show that for every $\varepsilon > 0$ there is a constant λ_ε, independent of P, with

$$H(P) \leqslant \lambda_\varepsilon R(ABC)^{3/2+\varepsilon}.$$

(1) Show that the absolute values of A, B, C, suitably ordered, are the coordinates of a rational point P' of the same line with $H(P') = H(P)$, and with the same radical of the product of the coordinates. So you may assume that $0 < A \leqslant B < C$. Show that then $ABC \geqslant H(P)^2/2$.[a]
(2) Let E be the curve (1.1). Show that $N_E \leqslant 2^7 R(ABC)$.[b]
(3) Let $\varepsilon > 0$. Use Conjecture 2.4 to show that $H(P)^4 \leqslant \mu_\varepsilon R(ABC)^{6+\varepsilon}$ for a constant μ_ε that depends only on ε, and conclude.

[a] *Hint:* two of A, B, $-C$ have the same sign.
[b] *Hint:* one of A, B, C is even.

3. From ABC to Szpiro

Problem 3.2 follows the proof of [Sil09, Proposition VIII.11.5(b)], which assumes that $\gcd(c_4, c_6) = 1$. The general case is [Sil09, Exercise 8.21]. The proof applies Conjecture 2.5 to the rational point $[c_4^3 : -c_6^2 : 1728\Delta]$ of the line $X + Y = Z$. The book has Δ in place of 1728Δ, a misprint corrected in its errata.

Proposition 3.1 (Weierstrass formulas [Sil09, §III.1]). *The invariants and the discriminant of a Weierstrass equation satisfy* $1728\Delta = c_4^3 - c_6^2$.

Problem 3.2 (ABC implies Szpiro).
Assume Conjecture 2.5. Let E be given by a global minimal Weierstrass equation, with invariants c_4, c_6 and discriminant $\Delta = \Delta_E$, and suppose that $\gcd(c_4, c_6) = 1$. You will show that for every $\delta > 0$ there is a constant κ_δ, independent of E, with

$$|\Delta_E| \leqslant \kappa_\delta N_E^{6+\delta}.$$

(1) Show that $c_4 c_6 \neq 0$, and that $P := [c_4^3 : -c_6^2 : 1728\Delta]$ is a rational point of the line $X + Y = Z$ in $\mathbb{P}^2$ with coprime integer coordinates. Deduce that

$$H(P) = \max\bigl(|c_4|^3, |c_6|^2, 1728|\Delta|\bigr).$$

(2) Show that $R(c_4^3 c_6^2 \cdot 1728\Delta) \leqslant 6|c_4 c_6| N_E$.[a]
(3) Let $0 < \varepsilon < 1/5$. Use Conjecture 2.5 and the inequalities $|c_4| \leqslant H(P)^{1/3}$ and $|c_6| \leqslant H(P)^{1/2}$ to show that

$$H(P)^{(1-5\varepsilon)/6} \leqslant 6^{1+\varepsilon} \lambda_\varepsilon N_E^{1+\varepsilon}.$$

Deduce the statement of the problem.

[a] *Hint:* every prime dividing Δ_E divides N_E.

References

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑

[Sil94] ———, *Advanced topics in the arithmetic of elliptic curves*, Graduate Texts in Mathematics, vol. 151, Springer-Verlag, New York, 1994. MR1312368 ↑